WORKSHEET 3.3 : EQUATION OF LINES IN $\mathbb{R}^3$

2a) // = same $\vec{m}$ $\therefore \vec{m} = (-4, 2, -5)$

b) $\vec{m} = (7,4,6) - (0,6,3)$

$= (7, -2, 3)$

c) // to $-x = \frac{y-3}{2} = \frac{z}{4}$.

$\Rightarrow \frac{x}{-1} = \frac{y-3}{2} = \frac{z}{4}$ $\therefore \vec{m} = (-1, 2, 4)$

3a) $\vec{r} = (1,1,2) + t(3,-1,-1)$

if $t=0$. $\vec{r} = (1,1,2)$

$t=1$ $\vec{r} = (4, 0, 1)$

b) $\left.\begin{array}{l} x = 4-2t \\ y = -2+5t \\ z = 5+4t \end{array}\right\} t \in \mathbb{R}$

if $t=0$, $x=4$, $y=-2$, $z=5$

$\therefore (4,-2,5)$

$t=1$, $x=2$, $y=3$, $z=9$.

$\therefore (2,3,9)$

c) $\frac{x-4}{3} = \frac{y+5}{4} = \frac{z+1}{-1}$

$\therefore \left.\begin{array}{l} x = 4+3t \\ y = -5+4t \\ z = -1-t \end{array}\right\} t \in \mathbb{R}$

if $t=0$, $x=4$, $y=-5$, $z=-1$

$(4,-5,-1)$

if $t=1$, $x=7$, $y=-1$, $z=-2$

$\therefore (7,-1,-2)$

4a) $\vec{r} = (2,4,6) + t(-1,-3,2)$, $t \in \mathbb{R}$

$\left.\begin{array}{l} x = 2-t \\ y = 4-3t \\ z = 6+2t \end{array}\right\} t \in \mathbb{R}$

$\frac{x-2}{-1} = \frac{y-4}{-3} = \frac{z-6}{2}$

b) $\vec{r} = (0,0,-5) + t(-1,4,1)$, $t \in \mathbb{R}$

$\left.\begin{array}{l} x = -t \\ y = 4t \\ z = -5+t \end{array}\right\} t \in \mathbb{R}$

$\frac{x}{-1} = \frac{y}{4} = \frac{z+5}{1}$

c) $\vec{r} = (1,0,0) + t(0,0,-1)$, $t \in \mathbb{R}$

$x = 1$
$y = 0$
$z = -t$
$\}$ $t \in \mathbb{R}$.

No symmetric Equations.
(not a single pair)

6a) $x = 2t$
$y = 3 + t$
$z = 1 + t$
$\}$ $t \in \mathbb{R}$

check $P(2,4,2)$ $\quad 2 = 2t \quad 4 = 3+t \quad 2 = 1+t$ ✓ $\quad \therefore P(2,4,2)$ is on the line
$\qquad\qquad 1 = t \quad\quad 1 = t \quad\quad 1 = t$

check $Q(-2,2,1)$ $\quad -2 = 2t \quad 2 = 3+t \quad 1 = 1+t$ $\quad \therefore Q(-2,2,1)$ NOT on the line
$\qquad\qquad -1 = t \quad\quad -1 = t \quad\quad 0 = t$ X.

Check $R(4,5,2)$ $\quad 4 = 2t \quad 5 = 3+t \quad 2 = 1+t$ $\quad \therefore R(4,5,2)$ NOT on the line.
$\qquad\qquad 2 = t \quad\quad 2 = t \quad\quad 1 = t$ X.

check $S(6,6,2)$ $\quad 6 = 2t \quad 6 = 3+t \quad 2 = 1+t$ X $\quad \therefore S(6,6,2)$ NOT on the line
$\qquad\qquad 3 = t \quad\quad 3 = t \quad\quad 1 = t$

b) Point $(a,b,-3)$ lies on the line
$\therefore \exists\, t$ s.t. it satisfies all components!

$a = 2t$
$a = 2(-4)$
$\boxed{a = -8}$

$b = 3 + t$
$= 3 + (-4)$
$\boxed{b = -1}$

$-3 = 1 + t$
$\boxed{-4 = t}$

$\therefore \boxed{\begin{matrix} a = -8 \\ b = -1 \end{matrix}}$.

7) $P_0(0,-1,1)$

M = midpoint.
$= \frac{1}{2}\left[(2,3,-2) + (4,-1,5)\right]$
$= \frac{1}{2}(6,2,3)$
$= (3,1,1.5)$

$\vec{m} = (3,1,1.5) - (0,-1,1)$
$= (3,2,0.5)$

$\therefore \boxed{\begin{matrix} x = 3t \\ y = -1 + 2t \\ z = 1 + 0.5t \end{matrix}\ \Big\}\ t \in \mathbb{R}}$

or $\vec{m}_2 = (6,4,1)$ $\quad \begin{matrix} x = 6s \\ y = -1 + 4s \\ z = 1 + s \end{matrix}\ \Big\}\ s \in \mathbb{R}$

8) $P_0(0,0,0)$ // through $(4,3,1)$ & $(-2,-4,3)$

$\therefore \vec{m} = (4,3,1) - (-2,-4,3)$

$= (6,7,-2)$

symmetric equations $\quad \frac{x-0}{6} = \frac{y-0}{7} = \frac{z-0}{-2}$

$$\boxed{\frac{x}{6} = \frac{y}{7} = \frac{-z}{2}}$$ ← simplified form.

9a) $\vec{r_1} = (1,0,3) + s(3,-6,3)$ $\qquad \vec{r_2} = (2,-2,5) + t(2,-4,2)$

$\vec{m_1} = (3,-6,3)$ $\qquad \vec{m_2} = (2,-4,2)$

$= 3(1,-2,1)$ $\qquad = 2(1,-2,1)$

Since $\vec{m_1} = \frac{3}{2}\vec{m_2}$ $\therefore$ collinear / parallel.

check $(1,0,3)$ in $\vec{r_2}$.

$1 = 2+2t$ $\qquad 0 = -2-4t$ $\qquad 3 = 5+2t$

$-1 = 2t$ $\qquad 2 = -4t$ $\qquad -2 = 2t$

$\frac{-1}{2} = t$ $\qquad \frac{-1}{2} = t$ $\qquad -1 = t$

different! $\therefore$ not coincident

$\therefore \vec{r_1} \neq \vec{r_2}$ are parallel but distinct

b) $\vec{r_1} = (2,-1,4) + s(3,0,6)$ $\qquad \vec{r_2} = (-3,0,1) + t(2,0,2)$

$\vec{m_1} = (3,0,6)$ $\qquad \vec{m_2} = (2,0,2)$

$= 3(1,0,2)$ $\qquad = 2(1,0,1)$

$\therefore \vec{m_1} \neq k\vec{m_2}$,

the two lines are neither coincident or parallel & distinct

c) $\vec{r_1} = (1,-1,1) + s(6,2,0)$ $\qquad$ $\vec{r_2} = (-5,-3,1) + t(-9,-3,0)$

$\vec{m_1} = (6,2,0)$ $\qquad$ $\vec{m_2} = (-9,-3,0)$

$= 2(3,1,0)$ $\qquad$ $= -3(3,1,0)$

$\therefore \vec{m_1} = -\frac{2}{3}\vec{m_2}$ $\qquad$ $\therefore$ collinear / parallel.

check $(1,-1,1)$ on $\vec{r_2}$:

$1 = -5 - 9t$ $\qquad$ $-1 = -3 - 3t$ $\qquad$ $1 = 1$ ✓

$6 = -9t$ $\qquad$ $2 = -3t$

$-\frac{2}{3} = t$ $\qquad$ $-\frac{2}{3} = t$

same t-value

$\therefore (1,-1,1)$ is on $\vec{r_2}$.

$\therefore$ coincident lines.